

Mini-screws / TADs: anchorage for tooth movement

A mini-screw is a small screw temporarily secured in the jawbone. It gives the orthodontist an anchor independent of the teeth and is removed once its task is complete.

What does TAD mean?

TAD stands for *Temporary Anchorage Device*. Mini-pin, mini-screw and orthodontic mini-implant are other names. Unlike an implant used for a crown, it does not carry a permanent dental replacement.

Direct anchorage

Force is transmitted from the screw through a suitable connection to the tooth or group of teeth being moved.

Indirect anchorage

The screw stabilises a group of teeth or an appliance. This then acts as the anchor for the planned movement of other teeth.

How can a mini-screw help?

Examples include closing gaps caused by congenitally missing teeth, uprighting tilted teeth, moving groups of teeth or moving a tooth further into the bone. It may also support alignment of impacted or displaced teeth. Any required surgical exposure and chain attachment are planned separately.

Where are screws placed?

In the palate, screws may be arranged sagittally, one behind another, or transversely, side by side, depending on the appliance. **In the upper and lower jaw**, suitable spaces between roots or other planned bone sites may also be used. Available space, bone, soft tissues and the intended force direction determine the choice.

The number and positions are selected individually. In growing jaws, developing teeth and growth must be considered. Not every site provides a safe insertion path.

Colour image sequence: an example of lower-jaw treatment

The screw provides anchorage; the orthodontist plans and controls tooth movement. If a tooth is fused to the bone, additional anchorage cannot overcome that lack of mobility.

Precise planning with CBCT and a surgical guide

In our practice this treatment is usually carried out with **three-dimensional planning and an individual surgical guide**. Entry point, direction and depth must be carefully coordinated, particularly in narrow spaces between roots.

1. **Review the findings and orthodontic plan.** We examine the teeth, soft tissues and hygiene, review existing images and agree the movement goal, screw position and appliance with the orthodontist.
2. **Select imaging for the specific question.** A CBCT scan is obtained when three-dimensional information is needed for safe planning. The necessity and available alternatives are assessed before every scan.
3. **Combine the data.** The 3D X-ray data are matched with a surface scan or model data. The insertion path is planned in relation to roots, developing teeth and other nearby structures.
4. **Make the guide.** The custom guide transfers the planned position to the mouth. Its correct seating is checked before insertion.

Radiation protection and limitations: CBCT is not automatically required for every mini-screw. It must offer a justified benefit for the individual treatment; this is especially important in children and adolescents. Even a correctly planned guide cannot entirely eliminate deviations or injuries.

[Read more about CBCT, radiation and the examination](#)

What happens during insertion?

Once local anaesthesia is effective, we place the screws at the planned sites. Depending on the bone and system, pilot drilling or a small soft-tissue opening may be needed. We check stability. If the guide does not seat securely or conditions differ from the plan, the procedure is adjusted or postponed.

Connection to the appliance and loading may begin immediately or later, according to the agreed plan and adequate stability. A prolonged unloaded healing phase, as used for many tooth-replacement implants, is not always needed.

Alternatives and treatment limitations

Depending on your findings, alternatives include anchorage on teeth, other orthodontic appliances, a different screw site or miniplates. An adjusted treatment goal or another way of managing a gap may be appropriate. Without extra anchorage, the intended result may be harder to achieve or require a different approach. We discuss these choices together.

Possible risks and further treatment

- **General surgical effects:** pain, swelling, bruising, bleeding, infection or delayed healing.
- **Soft tissues and comfort:** irritation of the gums, tongue or cheek, pressure sores or tissue growing over the screw head. A palatal appliance may initially affect speaking and eating.
- **Stability and material:** loosening, tilting, loss or fracture during insertion, use or removal. A replacement screw, different site or changed plan may be needed. If a fragment remains, removal versus retention with monitoring is considered individually.
- **Teeth and roots:** injury to neighbouring roots or developing teeth, damage to a tooth including loss of vitality and need for further treatment. A guide does not reduce this risk to zero.
- **Site-specific risks:** in the lower jaw, nerve injury may cause temporary or permanent altered sensation in the lip, chin or tongue. In the upper jaw or palate, blood vessels, the nasal floor or maxillary sinus may be affected; a mouth-to-sinus communication is possible at some sites.
- **Small components:** very rarely, a loose part may be swallowed or inhaled. Call 112 for breathing difficulty.
- **Medication and anaesthesia:** intolerance, allergy, circulatory or breathing problems are possible; severe reactions are rare.

How long does the screw stay?

For as long as it is needed for the planned tooth movement – often several months and longer for complex treatment. The orthodontist regularly checks movement, forces and connections. Surgical reviews take place as agreed and whenever a problem arises.

Removal and completion

Once no longer needed, the screw can usually be unscrewed; local anaesthesia is available according to the situation. A covered head or fractured part may need a larger opening. The small gum wound usually closes promptly. If stitches are needed, they are removed after 1–2 weeks.

Removing the screw does not necessarily mean that all orthodontic treatment is finished. The achieved tooth position often needs continued retention. Stable screws and a guide cannot guarantee successful tooth movement.

If the plan needs to change

If stability is insufficient, inflammation develops or movement fails to progress, we reassess the cause and alternatives. A screw may be removed or repositioned and the orthodontist adjusts the mechanics. Simply increasing force is not the answer. Orthodontic risks such as root shortening are also discussed by the orthodontist.

Anaesthesia and comfort options

Method	Description	Cost
Local anaesthesia Standard	You remain awake and feel pressure, not pain. If you feel pain, more anaesthetic is given immediately. You may use public transport home.	Statutory insurance benefit or included in the treatment package
Oral sedation (Tavor) Comfort option	A tablet about 60 minutes beforehand. It reduces stress without inducing sleep. Vital signs are monitored. An escort is needed; no driving for 24 hours. Collection from the practice is mandatory.	Approx. €80 Out-of-pocket
IV sedation Comfort option	A professional anaesthesia team provides medication at the sleep–wake boundary. You remain cooperative and usually have little memory. Recovery room and vital-sign monitoring afterwards. Tuesdays only. No food from 10 pm and no drinks from 6 am. Collection from the practice is mandatory.	Approx. €230 Out-of-pocket
General anaesthesia Comfort option	Deep sleep, with a breathing tube through the nose into the airway. No awareness of surgery. Recovery room and vital-sign monitoring afterwards. Tuesdays only. Same fasting instructions as IV sedation. Collection from the practice is mandatory.	Approx. €450 Out-of-pocket

Local anaesthesia is the standard and is sufficient for most patients. Sedation is an optional comfort service paid out of pocket. With every method, the anaesthetic is fully effective before we begin so that you do not feel pain.

For children and adolescents, the method is chosen individually according to age, health and cooperation. Tavor is not a routine recommendation for minors. Your agreed written fasting and medication instructions take priority. Medically necessary general anaesthesia and possible insurance coverage are assessed separately.

Preparation and individual arrangements

- Bring or arrange advance transfer of your referral, orthodontic plan, existing images, medication list and allergy details.
- Tell us about blood thinners, antiresorptive treatment, immunosuppression, bone-metabolism disorders and possible pregnancy. Do not stop prescribed medication yourself.
- For local anaesthesia alone, a light meal beforehand is usually appropriate. For sedation or general anaesthesia, follow your personal instructions and arrange collection and supervision. For at least 24 hours, do not drive, operate machinery, drink alcohol or make important decisions.
- For minors, clarify consent and accompaniment. Infection, inadequate hygiene or lack of suitable bone may require preliminary treatment or a different plan.

Costs: Screws and insertion, CBCT, planning and guide, orthodontic appliance, follow-up and removal, and optional comfort anaesthesia are itemised and agreed individually in advance. Insurance coverage is not guaranteed.

After insertion: cleaning and daily care

On the day: rest and, if needed, cool the area externally with breaks. Wait for numbness to wear off before eating; start with soft, lukewarm food. Avoid vigorous rinsing or touching the site. Initially avoid strenuous exercise; agree when to resume it with us.

During use: clean daily with a soft brush as demonstrated, including around the head and appliance. Do not smoke; use mouthwash or antibiotics only if prescribed. Do not turn, push or play with the screw using your tongue, and do not change traction forces yourself. Avoid pressing hard food against it.

Pain relief

Children and adolescents receive a **personal plan based on age and weight**. The following is the practice's adult regimen and applies only after individual approval. Medical conditions or other medication may require lower doses or a different drug.

Medicine	Dose if suitable	Note
Ibuprofen 600 mg	Every 6–8 hours, max. 3 doses/day	Take with food
Paracetamol 500–1000 mg	Alternatively/additionally every 6 hours	Max. 4 g/day; not suitable for all body weights
Novalgin drops	Up to 30 drops 4 times/day, only if prescribed	Only at 500 mg/ml and 20 drops/ml
No additional aspirin/ASA	Avoid as a painkiller	Do not independently stop prescribed long-term aspirin

If fever, chills, sore throat or painful mucosal sores develop while taking Novalgin/metamizole: **stop it immediately and seek urgent medical assessment**. Do not add paracetamol-containing combination products without advice.

Contact us immediately for warning signs

Bleeding that continues despite 30 minutes of pressure with a clean swab, fever, pus or clearly increasing pain and swelling: practice **06151 786540**.

After a procedure, outside practice hours: **+49 151 10437450**. If unavailable: emergency dental service **01805 60 70 11**.

For a life-threatening emergency, breathing difficulty, severe difficulty swallowing or rapidly increasing swelling of the floor of the mouth, neck or tongue: **call 112 immediately**.

Frequently asked questions

Can the screw replace a congenitally missing tooth?

No. It acts as a temporary anchor, for example to close a gap orthodontically or position teeth for a future restoration. A permanent implant crown is a different treatment.

Is a surgical guide always required?

We usually use individual 3D planning and a guide for these procedures. The method depends on the findings. The need for CBCT is assessed separately for the specific clinical question.

Can I speak, eat and brush normally?

A palatal appliance can feel unfamiliar at first. Once numbness has worn off, start with soft food. Clean accessible surfaces with a soft brush as demonstrated. Persistent pressure areas or discomfort should be checked.

What if the screw loosens or a part comes off?

Do not tighten, glue, cut or adjust it yourself. Contact our practice and the orthodontist promptly. Sudden coughing after a part comes loose requires assessment for possible inhalation; call 112 for breathing difficulty.

What if the tooth does not move?

The orthodontist checks progress, forces, space and attachments. Together we assess screw stability and whether the plan needs changing. A tooth fused to the bone cannot be moved normally even with additional anchorage.

Your treatment checklist

Before starting, check: movement goal and screw site understood? CBCT indication and guide discussed? Personal medication instructions provided? Review and orthodontic follow-up arranged? If sutures are placed, removal after 1–2 weeks arranged? Contact route for looseness known?

Clinical background

Clinical background: MKG im Carree practice protocol; [Gebrauchsanweisung zur temporären skelettalen Verankerung / temporary anchorage instructions \(05/2021\)](#); [British Orthodontic Society: Orthodontic mini-screws](#); [NICE HTG152: orthodontic anchorage \(Empfehlungen / recommendations 2007\)](#); [AWMF 083-005: Dentale digitale Volumentomographie](#). Accessed 09 September 2026. This information supplements your personal consent discussion.